Name: ______________________________

Class: ______________________________

Date: ______________________________

Mr. Rawson • GCSE Physics

Energy 2 — Kinetic Energy

Key concept

$$E_k = \tfrac{1}{2}\, m\, v^2$$

E_k	kinetic energy	joules, J
m	mass	kilograms, kg
v	speed	metres per second, m/s

You must recall this equation. It is not on the equation sheet.

Worked example

A cyclist of total mass $75\,\text{kg}$ travels at $8.0\,\text{m/s}$. Calculate the energy in their kinetic store.

$E_k = \frac{1}{2}mv^2 = \frac{1}{2} \times 75 \times 8.0^2 = \frac{1}{2} \times 75 \times 64 = 2400\,\text{J}$

Square the speed first. $8.0^2 = 64$, *not* 16.

1) **Calculate** the kinetic store in each case. Show every step.

a) A $1200\,\text{kg}$ car travelling at $15\,\text{m/s}$. **[3]**

b) A $0.045\,\text{kg}$ golf ball leaving the tee at $60\,\text{m/s}$. **[3]**

c) *The same car as part (a)*, now travelling at $30\,\text{m/s}$. Compare your answer with (a). **[3]**

Name: ______________________________ **Date:** ______________________

2) **Complete** the table. The mass is the same every time — only the speed changes.

Speed / m/s	E_k / J	**How many times bigger than the row above?**
10		—
20		
30		
40		

Use a mass of $1200\,\mathrm{kg}$ *throughout.*

3) State** what happens to the kinetic store when the speed doubles, and explain why it is not simply twice as big.* ***[2]

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4) Calculate** the kinetic store of a* $900\,\mathrm{kg}$ *car travelling at* $72\,\mathrm{km/h}$. ***[4]

***Careful:** The equation needs speed in metres per second. Convert first, and show the conversion — it earns a mark.*

5) Determine** the speed of a* $1500\,\mathrm{kg}$ *car whose kinetic store is* $300\,000\,\mathrm{J}$. ***[4]